

Switched-Tensor Lookup Certification for Wide-Word Integer Evaluation over Binary Fields

A working note

F2Z project notes

2026-07-20 (v0.1 — design note; costs are modeled, not measured)

Abstract

We consider polynomial commitment schemes for W -bit integer data committed bit-wise over a characteristic-two field, where an evaluation claim $y = \text{MLE}[\text{INT}(D)](r) \bmod q$ is reduced to weighted integer folds carried in the exponent of $\mathbb{K} = \mathbb{F}_{2^{128}}$ and certified by grand-product GKR (“forest”) arguments. The deployed certification costs one bit-affine leaf per committed bit and is therefore blind to the word width W : at equal total bit count, wide-word instances cannot outperform $W = 1$ instances. We describe a *switched tensor order* for the per-leaf weights $w_b \cdot 2^j$: the narrow place-value factors 2^j are moved to the outer (aggregation) axis, where they are handled in the clear, and the wide ~ 100 -bit evaluation weights w_b are moved into *public* lookup tables shared by all W bit-planes. The sharing shrinks the table side of the lookup argument by a factor of exactly W , which is the term that made previous lookup-based designs uncompetitive. Queries are bound by a characteristic-two-sound certificate; we give two instantiations, a multiplicative grand-product multiset argument built entirely from the deployed forest machinery, and a multiplicity-free logarithmic-derivative (pushforward) argument. We prove soundness of the multiset instantiation, give a cost model placing the prover at ≈ 0.8 – $1.05\times$ the per-bit baseline (i.e. a modeled speedup of ≈ 0.95 – $1.3\times$, against a required decision bar of $1.5\times$), and specify the measurement gate that must adjudicate the design. A by-product of the construction is that weight chunking disappears entirely: the wide weights never meet the exponent-capacity bound.

1 Introduction

Setting. A prover commits to a bit tensor $D \in \{0, 1\}^{2^t \times W \times 2^s}$, $W = 2^w$, interpreted as a $2^t \times 2^s$ matrix of W -bit integers $\text{INT}(D)(b, c) = \sum_{j < W} 2^j D(b, j, c)$, for a total of $N = 2^{t+w+s}$ committed bits. The commitment is a linear-map-friendly PCS over $\mathbb{F}_2$ (in the reference implementation: a ring-switched, recursively composed proximity-tested code) whose openings discharge multilinear evaluation claims about the committed bits over $\mathbb{K} = \mathbb{F}_{2^{128}}$. The statement is a wide-modulus evaluation

$$y = \sum_{b, c} \text{eq}(b, r_1) \text{eq}(c, r_2) \cdot \text{INT}(D)(b, c) \pmod{q}, \quad q = 2^{100} - 15.$$

Lifting the row weights to canonical representatives $w_b := \text{eq}(b, r_1) \bmod q \in [0, q)$ and keeping the column weights $w'_c := \text{eq}(c, r_2) \in \mathbb{F}_q$ at full width, the claim factors through the per-column

integer folds

$$u_c = \sum_b w_b \cdot \text{INT}(D)(b, c) \in \mathbb{Z}_{\geq 0}, \quad y = \sum_c w'_c u_c \pmod{q}. \quad (1)$$

Characteristic two prohibits certifying (1) by a sumcheck over $\mathbb{K}$ (integers lifted into $\mathbb{K}$ collapse to parity), so the fold rides *multiplicatively*: for a generator α of $\mathbb{K}^\times$ (of order $\text{ord}(\alpha) = 2^{128} - 1$), the map $n \mapsto \alpha^n$ is injective on $[0, 2^{128} - 1)$, and the prover certifies α^{u_c} as a grand product of *bit-affine leaves*

$$\ell(b, j) = 1 + D(b, j, c) \cdot (\alpha^{w_b 2^j} - 1) = (\alpha^{w_b 2^j})^{D(b, j, c)},$$

one per committed bit, by a lazy merged GKR product forest. The bases $\alpha^{w_b 2^j}$ are *public*; the leaf-layer claim is linear in the committed bits and discharges through the PCS opening at no extra certification cost. Because u_c may exceed the exponent capacity, the weights are chunked into c_w -bit limbs with $c_w = 127 - t - W$ and the protocol repeats per limb; this detail is orthogonal here and, as we will see, vanishes in the present construction.

The question. The prover cost of the baseline is $\Theta(N)$ multiplications in $\mathbb{K}$ — one bit-affine leaf, with its share of sumcheck rounds, per committed bit — with a measured constant of $c_{\text{bit}} \approx 10.5 \text{ ns}$ per bit on the reference hardware. This cost is blind to W . Wide words carry visible structure (all W bits of a cell share one row weight; within a cell, the bases form a Frobenius orbit $\alpha^{w_b 2^j} = \varphi^j(\alpha^{w_b})$), and the question is whether any of it can push the prover *below* one leaf per bit. Prior attempts in this project — an octet-grouped lookup at $W = 1$, a wide-outer tensor decomposition, base- 2^k weight slicing — were all measured or adjudicated at $\leq 1.0\times$; a recurring accounting, which we record as Remark 2.2, explains why: certified work is paid per *private* exponent bit, and only public-base precomputation absorbs weight width for free.

Contribution. We observe that the per-leaf weight $w_b \cdot 2^j$ is a tensor with a narrow factor (2^j , one nonzero bit) and a wide factor (w_b , ~ 100 bits), and that all prior designs placed the wide factor on the private (certified) axis. Switching the tensor order places:

- the *wide* factors inside public lookup tables $T_\beta[a] = \alpha^{\langle a, w_\beta \rangle}$ indexed by octets of committed bits — the width-free position — and, decisively, *shared by all W bit-planes and all columns*, shrinking the lookup argument's table side by a factor of W ;
- the *narrow* factors 2^j on the outer aggregation axis, where they are handled in the clear (or by a companion commitment), never inside a certified exponent.

Sections 3–5 give the construction and its soundness; Section 6 prices it honestly against the measured baseline and specifies the measurement gate. We emphasize up front that this note argues *soundness*, not *profit*: the cost model straddles the project's $1.5\times$ adoption bar, and four predecessor designs died at measurement.

2 Preliminaries

Notation. $\mathbb{K} = \mathbb{F}_{2^{128}}$ with Frobenius $\varphi(x) = x^2$; φ is a field automorphism fixing $\mathbb{F}_2$, of order 128, so $\varphi^{-k} = \varphi^{128-k}$ is locally computable. $B = \{0, 1\}$. For $x \in B^n$, $\text{eq}(x, r) = \prod_i (x_i r_i + (1 -$

$x_i)(1 - r_i)$). For $f : B^n \rightarrow \mathbb{K}$, $\widehat{f}$ denotes its multilinear extension and $\widehat{f}(r) = \sum_x \text{eq}(x, r) f(x)$. α is a fixed verifier-checked generator of $\mathbb{K}^\times$. All committed vectors are $\mathbb{F}_2$ -valued; the PCS discharges claims of the form $\sum_x \lambda(x) M(x)$ for public $\mathbb{K}$ -valued λ and committed bits M (“linear read-offs”), and batched multilinear evaluations of M at transcript-derived points.

We use the merged product forest as a black box: given per-tree leaf vectors over $\mathbb{K}$, it certifies all tree roots with prover cost proportional to the number of leaves, at measured per-leaf constants c_{bit} (bit-affine leaves, public bases, free linear discharge) and $c_{\text{dense}} \approx 2c_{\text{bit}}$ (dense prover-supplied leaf values), plus a leaf-layer evaluation claim that the caller must discharge.

Lemma 2.1 (Twisted evaluation). *Let $f : B^n \rightarrow \mathbb{K}$ and let $f^{(2)}(x) = f(x)^2$ pointwise. Then for every $r \in \mathbb{K}^n$,*

$$\widehat{f^{(2)}}(r) = \left(\widehat{f}(\varphi^{-1}(r)) \right)^2,$$

where φ^{-1} is applied coordinate-wise. Iterating, $\widehat{f^{(2^k)}}(r) = \varphi^k(\widehat{f}(\varphi^{-k}(r)))$.

Proof. Squaring is additive in characteristic two, so $\widehat{f}(s)^2 = \sum_x \text{eq}(x, s)^2 f(x)^2$. For $x_i \in \{0, 1\}$ each factor of eq squares to $(x_i s_i + (1 - x_i)(1 - s_i))^2 = x_i s_i^2 + (1 - x_i)(1 - s_i)^2 = x_i s_i^2 + (1 - x_i)(1 - s_i^2)$, using $x_i^2 = x_i$ and $(1 - s)^2 = 1 - s^2$ in characteristic two. Hence $\text{eq}(x, s)^2 = \text{eq}(x, s^{(2)})$ with $s^{(2)} = (s_i^2)_i$, and substituting $s = \varphi^{-1}(r)$ gives the claim. $\square$

Remark 2.2 (Width conservation, informal). The statement (1) involves $q_{\text{bits}} \cdot N$ weight-bit $\times$ data-bit products. A certified object whose private input is a c -bit committed index accounts for at most c committed bits per query, while a *public* table entry may absorb arbitrarily many weight bits at no certified cost. Consequently: private-base exponentiation pays one certified unit per exponent bit; public-base precomputation is width-free. The baseline leaf is the $c = 1$ endpoint (one committed bit selecting one public constant, certificate free by linearity). The present construction is the $c = 8$ point of the same family: eight committed bits select one of 256 public constants, and the price of leaving the endpoint is a membership certificate. Everything in Section 6 is an accounting of that certificate.

3 The switched tensor order

Fix the octet parameter $n_2 = 8$ (any $n_2 \leq 16$ works; $n_2 = 8$ is the knee in our model). Assume $t \geq 3$ and partition the row axis into octets: for $\beta \in [2^{t-3}]$ let $w_\beta = (w_{8\beta}, \dots, w_{8\beta+7})$ and, for plane $j \in [W]$ and column c , let

$$a_{j,\beta,c} = (D(8\beta + i, j, c))_{i < 8} \in B^8$$

be the corresponding committed octet.

3.1 Public tables and the sharing factor

Definition 3.1 (Row-octet tables). For each β , define the public table $T_\beta : B^8 \rightarrow \mathbb{K}^\times$,

$$T_\beta[a] = \alpha^{\langle a, w_\beta \rangle} = \prod_{i < 8} (\alpha^{w_{8\beta+i}})^{a_i}.$$

There are $|E| = 2^{t-3} \cdot 2^8 = 2^{t+5}$ table entries in total, built incrementally in one $\mathbb{K}$ -multiplication per entry from the 2^t bases α^{w_b} (which both parties already compute in the baseline). Two properties are load-bearing:

(i) *Sharing*. T_β depends only on the row octet — not on the plane j or the column c . All $W \cdot 2^s$ (plane, column) pairs query the same 2^{t+5} entries. A $W = 1$ instance of the same total size has 2^{t+w} rows with pairwise distinct weights and hence 2^{t+w+5} entries: the table size is a factor W larger. This factor is the entire advantage of wide words, and it lands exactly on the terms (table construction, multiplicity bookkeeping, table-side certification) that dominated and killed the prior octet-grouped design at $W = 1$.

(ii) *Product form*. T_β is multilinear-with-multiplicative- structure in its index: for any $\rho \in \mathbb{K}^8$,

$$\sum_{a \in B^8} \text{eq}(a, \rho) T_\beta[a] = \prod_{i < 8} \left((1 - \rho_i) + \rho_i \alpha^{w_{8\beta+i}} \right), \quad (2)$$

so the verifier evaluates the table's multilinear extension at any point in $O(2^{t-3} \cdot 8)$ multiplications given the α^{w_b} — the table is never committed and never materialized by the verifier.

3.2 Per-plane folds; chunking disappears

Write the statement with the place values outermost:

$$u_c = \sum_{j < W} 2^j v_{j,c}, \quad v_{j,c} := \sum_{\beta} \langle a_{j,\beta,c}, w_\beta \rangle = \sum_b w_b D(b, j, c). \quad (3)$$

Lemma 3.2 (Magnitude). $0 \leq v_{j,c} < 2^t q < 2^{t+100}$. In particular, for $t \leq 27$ every per-plane fold is individually below $2^{127} < \text{ord}(\alpha)$, and the map $v \mapsto \alpha^v$ is injective on the occurring values.

The protocol therefore certifies the *per-plane* folds $v_{j,c}$ multiplicatively and performs the outer aggregation $u_c = \sum_j 2^j v_{j,c}$ and the read-off $y = \sum_c w'_c u_c \bmod q$ entirely in $\mathbb{F}_q$, on the clear (or committed, §4.4) values $v_{j,c}$. Two consequences:

- **No weight chunking**. The wide weights live whole inside the public table entries; no certified exponent ever multiplies a weight by a cell value. The baseline's limb decomposition ($c_w = 127 - t - W$, $L = \lceil q_{\text{bits}}/c_w \rceil$ instances) is gone for every $t \leq 27$ and every W — in particular the forced $L = 2$ of the baseline at $W = 32$.
- **Payload**. There are $W \cdot 2^s$ per-plane folds instead of 2^s per-column folds. Sent in the clear they cost $16 W 2^s$ bytes (1 MiB at $W = 32$, $s = 11$) and $W 2^s$ verifier exponentiations; at scale the committed-fold companion of §4.4 is mandatory.

Remark 3.3 (Why the plane axis is not folded in the exponent). It is tempting to aggregate $\sum_j 2^j v_{j,c}$ inside the forest by Frobenius-twisted combines $x \cdot \varphi^{2^k}(y) \cdot C^\kappa$ with $C = \alpha^{-q}$ and committed correction digits κ , keeping intermediate exponents reduced modulo q . This is *unsound as stated*: correction digits are constrained only by their committed widths, the representable corrections span far more than 128 bits of freedom, and $\text{ord}(\alpha) = 2^{128} - 1 \not\equiv 0 \pmod{q}$ — so a malicious prover can grind κ to wrap the exponent around $\text{ord}(\alpha)$ and shift the final value by a nonzero multiple of $\text{ord}(\alpha) \bmod q$, changing the read-off. Repairing this requires range certificates on every intermediate reduced exponent, at which point the design is strictly worse than keeping the plane axis in the clear. The per-plane formulation (3) avoids the entire issue; no Frobenius twist appears anywhere in the prover's circuit.

3.3 The value forests

For each plane–column pair (j, c) the prover runs a product tree of depth $t - 3$ over the octet values

$$V_{j,c} = \prod_{\beta} v_{j,\beta,c}^{\text{leaf}}, \quad v_{j,\beta,c}^{\text{leaf}} \stackrel{!}{=} T_{\beta}[a_{j,\beta,c}],$$

so that, for an honest prover, $V_{j,c} = \alpha^{v_{j,c}}$ (Lemma 3.2 bounds the exponent). All $W2^s$ trees run in one merged forest call. The leaves are *prover-supplied* table reads — $N/8$ dense leaves in place of N bit-affine leaves — and the entire soundness burden moves to certifying the marked equality $\stackrel{!}{=}$, i.e. that every supplied leaf equals the public table entry at the committed index. That certificate is the subject of the next section.

4 The certification layer

4.1 Fingerprints and multiplicities

After the commitments are fixed (see §4.3 for the transcript order), the verifier draws $\gamma_{\text{pos}} \in \mathbb{K}^{t-3}$, $\gamma_{\text{idx}} \in \mathbb{K}^8$, $\mu, \delta \in \mathbb{K}$ and both parties define the fingerprint of a (position, index, value) triple:

$$\text{fp}(\beta, a, v) = \langle \gamma_{\text{pos}}, \text{bits}(\beta) \rangle + \langle \gamma_{\text{idx}}, a \rangle + \mu v \in \mathbb{K}.$$

Let $Q = \{(j, \beta, c)\}$ be the query set, $|Q| = N/8$, with query triple $\tau_q = (\beta_q, a_q, v_q)$, and let $E = \{(\beta, a)\}$ be the table domain, $|E| = 2^{t+5}$, with table triple $\tau_e = (\beta, a, T_{\beta}[a])$. The prover counts multiplicities

$$m_e = \#\{q \in Q : (\beta_q, a_q) = e\} \leq W2^s,$$

and commits them in binary as $s+w+1$ extra bit-planes over E ($\approx (s+w+1)2^{t+5}$ bits, i.e. 3–5% of N at $s \in [9, 11]$). Crucially m is a function of D alone — no challenge enters it — so the multiplicity planes are part of the *same* first-round commitment as the data.

4.2 The multiplicative grand-product multiset argument

The lookup relation is certified by the polynomial identity

$$\prod_{q \in Q} (\delta - \text{fp}(\tau_q)) = \prod_{e \in E} (\delta - \text{fp}(\tau_e))^{m_e}, \quad (4)$$

with both sides certified as grand products:

Witness side. The factors $\delta - \text{fp}(\tau_q)$ are grouped by (j, c) into $W2^s$ trees of depth $t - 3$ — the same shape as the value forests — and run as a second claim family in the *same* merged forest call. The leaf at query q is $G_{\beta}[a_q]$ where $G_{\beta}[a] = \delta - \langle \gamma_{\text{pos}}, \text{bits}(\beta) \rangle - \langle \gamma_{\text{idx}}, a \rangle - \mu T_{\beta}[a]$ is another public per-octet 256-entry table (built from T_{β} by 256 additions).

Table side. Decomposing $m_e = \sum_k 2^k m_{e,k}$,

$$\prod_e (\delta - \text{fp}(\tau_e))^{m_e} = \prod_e \prod_{k \leq s+w} \underbrace{\left(\varphi^k(\delta - \text{fp}(\tau_e)) \right)}_{B_{e,k}}^{m_{e,k}} = \prod_{e,k} (1 + m_{e,k} (B_{e,k} - 1)),$$

which is a forest of *bit-affine leaves with public bases* over the committed multiplicity bits — exactly the baseline leaf shape, with its free linear discharge, over $(s+w+1) 2^{t+5}$ leaves. The verifier’s exit evaluations of the base family use the product form (2) together with Lemma 2.1: $\sum_e \text{eq}(e, \rho) \varphi^k(\delta - \text{fp}(\tau_e)) = \varphi^k(X(\varphi^{-k}(\rho)))$ where $X(\sigma) = \delta - (\text{linear terms}) - \mu \hat{T}(\sigma)$ is closed form. The table is never committed.

The shared byte-driven discharge. The two dense claim families (values $T_\beta[a_q]$ and fingerprints $G_\beta[a_q]$) exit the merged forest at one shared point with leaf-MLE claims that are both affine in the same two residuals: the linear read-offs of the position/index bits (free) and the evaluation $\hat{v}(\rho)$ of the table-read vector. That residual is discharged by *byte-driven case-LUT rounds*: the leaf value at each query position is one gather from a public 256-case table keyed by the committed octet, and eight bit-driven sumcheck rounds bind the octet bits against progressively folded case tables ($256 \rightarrow 128 \rightarrow \dots$), after which the remaining claim is a plain multilinear evaluation of the committed bits, batched into the PCS opening. This is the depth-8 extension of the deployed depth-2/3 bit-driven leaf rounds; the project’s byte-table measurements support pricing these rounds at dense-round parity. This discharge is what closes the “free wires” loophole: leaf values are never free prover inputs; they are, verifiably, functions of the committed bits through public tables fixed before δ .

4.3 Transcript order

(1) Commit D together with the multiplicity planes m (both challenge-independent). (2) The claim (r, y) is fixed; weights w_b, w'_c are public functions of r . (3) [Committed-fold variant only: commit the fold bits, §4.4.] (4) Draw $\gamma_{\text{pos}}, \gamma_{\text{idx}}, \mu, \delta$. (5) Run the merged forests (value + fingerprint families; multiplicity forest), the shared discharge, and the openings. (6) Verifier: rebuild α^{w_b} (as in the baseline), check the multiset root identity (4), check V -roots against the per-plane folds (exponentiation in the clear variant; the linkage of §4.4 otherwise), range-check the folds, and compute $y = \sum_c w'_c \sum_j 2^j v_{j,c} \bmod q$.

4.4 Committed folds (companion gadget)

At scale the $W2^s$ clear folds are replaced by a committed bit-decomposition bound by a static exponentiation forest: the fold bits (at most $t+100$ planes per fold; the bit width itself enforces the range bound) are committed after step (2); tiny bit-affine trees with the public Frobenius base chain $\alpha^{2^k} = \varphi^k(\alpha)$ certify $\alpha^{v_{j,c}}$; a shared-challenge equality of root-MLE claims links them to the value forests’ roots without ever sending roots; and the $\mathbb{F}_q$ read-off becomes a single-column instance of the mod- q protocol itself over the fold bits, terminating in one clear 128-bit integer. At $s \geq 10$, $W = 32$, the fold-bit instance is large enough to sit inside the audited parameter regime of the underlying opener. We omit details; the gadget is specified in the project’s u-suppression note and is independent of the present construction.

4.5 Instantiation B: the pushforward (logup*) certificate

The fingerprint and multiplicity forests can be replaced by a logarithmic-derivative argument in its indexed, multiplicity-free (pushforward) form: query terms eq-weighted as $1/(\delta - \text{fp}(\tau_q))$, table terms $Y_e/(\delta - \text{fp}(\tau_e))$ with Y the pushforward of the query weights along the index map, certified by a fraction-addition GKR. Characteristic-two soundness of this form is established in the project’s route-(b) certificate (`quotient_gkr.rs`), with the fraction layer’s per-term cost measured at $\phi \approx 1.25$ – 1.5 product-forest terms. Its distinguishing adder is that Y ($|E|$ field elements) is prover-supplied and must be bound — a second commitment of $\approx 2^{7-s}N$ bits — whereas the multiset instantiation commits only the 3–5% multiplicity planes and reuses the deployed forest machinery wholesale. The model of Section 6 slightly favors the multiset arm at $s \leq 10$ and the pushforward arm at $s \geq 11$; both should be gated in the same harness, since they share the value forests, tables, and discharge.

5 Soundness

Lemma 5.1 (Fingerprint injectivity). *Fix any two distinct triples $\tau \neq \tau'$ occurring among $\{\tau_q\} \cup \{\tau_e\}$. Then $\Pr_{\gamma, \mu}[\text{fp}(\tau) = \text{fp}(\tau')] \leq 1/|\mathbb{K}|$, and by a union bound all occurring triples receive distinct fingerprints except with probability at most $(|Q| + |E|)^2/|\mathbb{K}|$.*

Proof. $\text{fp}(\tau) - \text{fp}(\tau')$ is a nonzero $\mathbb{K}$ -linear form in $(\gamma_{\text{pos}}, \gamma_{\text{idx}}, \mu)$: if the position or index bits differ it is nonconstant in the corresponding γ coordinate; otherwise $v \neq v'$ and it equals $\mu(v' - v)$. $\square$

Lemma 5.2 (Multiset binding). *Condition on distinct fingerprints (Lemma 5.1). If the multiset $\{\{\tau_q\}\}_{q \in Q}$ differs from $\biguplus_e \{\tau_e \text{ with multiplicity } m_e\}$, then the two sides of (4), as polynomials in δ , are distinct, and they agree at a random δ with probability at most $\max(|Q|, \sum_e m_e)/|\mathbb{K}| \leq 2^{t+s+w+6}/2^{128}$.*

Proof. Both sides are monic products of linear factors; by unique factorization in $\mathbb{K}[\delta]$ they are equal iff their root multisets coincide with identical *integer* multiplicities. Distinct fingerprints make the root multisets faithful images of the triple multisets. Schwartz–Zippel gives the agreement bound. Note the multiplicities are counted in $\mathbb{Z}$ as factor multiplicities — this is the point at which the argument is sound in characteristic two, where the additive (logarithmic-derivative-with-field-multiplicities) form $\sum_e m_e/(\delta - \text{fp}_e)$ would collapse even counts. $\square$

Theorem 5.3 (Soundness, sketch). *Under the transcript order of §4.3, if the verifier accepts then, except with probability $O((N^2 + \text{rounds})/|\mathbb{K}|)$ over the challenges plus the binding errors of the commitment and opening, the committed data D satisfies $y = \text{MLE}[\text{INT}(D)](r) \bmod q$.*

Proof sketch. (i) The forest GKRs pin the claimed roots to their actual leaf wires (standard grand-product GKR soundness over the merged layers). (ii) The byte-driven discharge pins each leaf wire of both dense families to the public table read at the committed octet: the case-LUT rounds are exact algebraic identities over the committed bits, and their terminal claims are discharged by the PCS opening; hence the supplied values satisfy $v_q = T_{\beta_q}[a_q]$ provided the multiset identity holds — and independently the fingerprint leaves equal $G_{\beta_q}[a_q]$ as functions of the same bits, so no adaptive choice of wires survives the discharge. (iii) By Lemmas 5.1–5.2, acceptance of (4) forces the query multiset to equal the table multiset, i.e.

every query’s value is the table entry at its committed index. (iv) Therefore each value tree’s root equals $\alpha^{v_{j,c}}$ with $v_{j,c}$ the true fold of (3); by Lemma 3.2 and the range checks, the clear (or committed) folds equal the true folds by injectivity of $v \mapsto \alpha^v$ below $\text{ord}(\alpha)$. (v) The read-off $y = \sum_c w'_c \sum_j 2^j v_{j,c} \bmod q$ is deterministic public arithmetic on bound integers, and equals $\text{MLE}[\text{INT}(D)](r) \bmod q$ by (1)–(3). $\square$

6 Cost model and the measurement gate

All entries are in units of $c_{\text{bit}} \cdot N$ (the measured baseline: one bit-affine leaf per committed bit, $c_{\text{bit}} \approx 10.5 \text{ ns}$ on the reference hardware; $c_{\text{dense}} \approx 2c_{\text{bit}}$ measured). The baseline column is the deployed forest *with* the per-leaf chunking fix, i.e. $L = 1$: the fair comparison point, equal to a $W = 1$ instance of the same size.

component	count	model cost
value forests	$N/8$ dense leaves	0.25
fingerprint forests	$N/8$ dense leaves	0.25
shared byte discharge	8 rounds over $N/8$	0.25–0.5
multiplicity forest	$(s+w+1)2^{t+5}$ bit-affine	0.03–0.05
tables T, G	2^{t+5} mults + 256-adds	< 0.01
read-off / linkage	§4.4	≤ 0.03
total		0.78–1.05

Model speedup vs. the baseline: $\approx 0.95\text{--}1.3\times$, with the discharge constant the widest uncertainty. For the pushforward arm, replace (fingerprint + multiplicity) by fraction terms at measured $\phi \approx 1.3$ and add the Y -binding instance ($\approx 2^{7-s}N$ committed bits). Three observations:

(a) *Where the W factor acts.* The prior $W=1$ octet design paid a table-side factor $c_f = 1 + 2^{8-s}$ inside its certificate (measured fatal at $s=7$, marginal at $s=10$); here the sharing reduces the analogous terms to the multiplicity forest and table rows above — $O((s+w)/2^s)$ — because the table population is $W\times$ smaller relative to the queries. This is the only place wide-word structure does real work, and it is worth exactly $\Theta(W)$ on those terms.

(b) *$W = 1$ non-applicability.* At $W = 1$ there are no planes to share across; the construction degenerates to the prior design and its measured verdict. The advantage claimed here *exists only for wide cells* — consistently with the project’s earlier finding that no arrangement makes W -blind certification cheaper.

(c) *The bar.* The project’s adoption bar for certificate-bearing redesigns is a measured $\geq 1.5\times$ prover advantage with all adders priced (extra commitments, proof bytes, verifier work, peak memory), a bar set by the history of four predecessor designs whose op-count models overstated measured performance — “the witness forest cancels the win” was measured four times. The model above *straddles* 1.0; this note therefore specifies, and does not prejudge, the gate:

- **Harness:** the existing octet-gate harness extended with (i) shared tables across planes, (ii) both certificate arms measured same-run, (iii) the multiplicity planes and (for arm B) the Y instance charged as real committed data, (iv) the byte discharge implemented, not modeled.

- Shapes: genuine $W=32$ at $(t, s) \in \{(12, 9), (12, 11)\}$, run against the current fused, bit-driven, $L=1$ baseline on an idle box, medians of ≥ 3 , one shape per process.
- Decision: adopt only if the gate clears $1.5\times$ with adders; record the outcome either way in the project ledger.

7 Related constructions

Logarithmic-derivative lookups originate with LogUp [1] and its GKR formulation [2]; the multiplicity-free indexed variant is logup* [3]. In characteristic two the additive form is unsound (even multiplicities vanish); the two sound routes used here are the multiplicative grand-product multiset (folklore; in-project reference implementation in the `f2-clean-lookup` machinery) and the pushforward form (in-project `quotient_gkr.rs`). Binius [4] certifies exponentiation by committed bits with the same bit-affine product structure; its blueprint’s exponentiation gadget [5] matches our baseline in the static-base case (public base, per-exponent-bit cost, “no windowing”), and its Frobenius phase — untwisting V^{2^i} claims through φ^{-i} -twisted evaluation points — is Lemma 2.1, which we need only on the verifier side (table-side exit evaluations), having removed all in-circuit twists by Remark 3.3. The width-conservation accounting (Remark 2.2) and the measured verdicts cited throughout are from the project ledger (f2-int workstream, 2026-06/07): the $W=1$ octet gate, the wide-outer tensor and weight-slicing adjudications, and the kernel-parity re-gate of the pushforward certificate.

8 Status

This is a design note. Soundness (Section 5) we consider settled up to the standard forest/opening arguments it inherits; performance is a model straddling the adoption bar, to be settled by the gate of Section 6 and by nothing else. Independent of the gate’s outcome, two elements of the analysis stand on their own: the per-plane formulation removes weight chunking for every $t \leq 27$ (subsuming the baseline’s $L=2$ penalty at $W=32$), and Remark 3.3 documents a soundness pitfall (correction- digit wraparound) that any in-exponent aggregation design must avoid.

References

- [1] U. Haböck. *Multivariate lookups based on logarithmic derivatives*. ePrint 2022/1530.
- [2] S. Papini, U. Haböck. *Improving logarithmic derivative lookups using GKR*. ePrint 2023/1284.
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- [4] B. E. Diamond, J. Posen. *Succinct arguments over towers of binary fields*. ePrint 2023/1784.
- [5] Irreducible. *Binius blueprint: exponentiating multilinear*. <https://www.binius.xyz/blueprint/backend/muls/exponentiating>.